

Aluminum production

Aluminum cannot be reduced easily like iron. It is too reactive so there are no suitable reducing agents. Therefore this extremely useful metal is extracted from its ore—generally bauxite (Al_2O_3)—by electrolysis (see page 148). The ore is dissolved in molten cryolite (a mineral compound of sodium, aluminum, and fluorine). This electrolyte (liquid that can conduct electricity) is more than $1,000^\circ\text{C}$ ($1,832^\circ\text{F}$) and is held in a tank or cell, lined with graphite carbon. This lining acts as the negatively charged cathode, while more graphite blocks are used as the positively charged anodes.

Before the invention of electrolysis in the 1880s, pure **aluminum** was more expensive than gold.

▷ The Hall-Héroult process

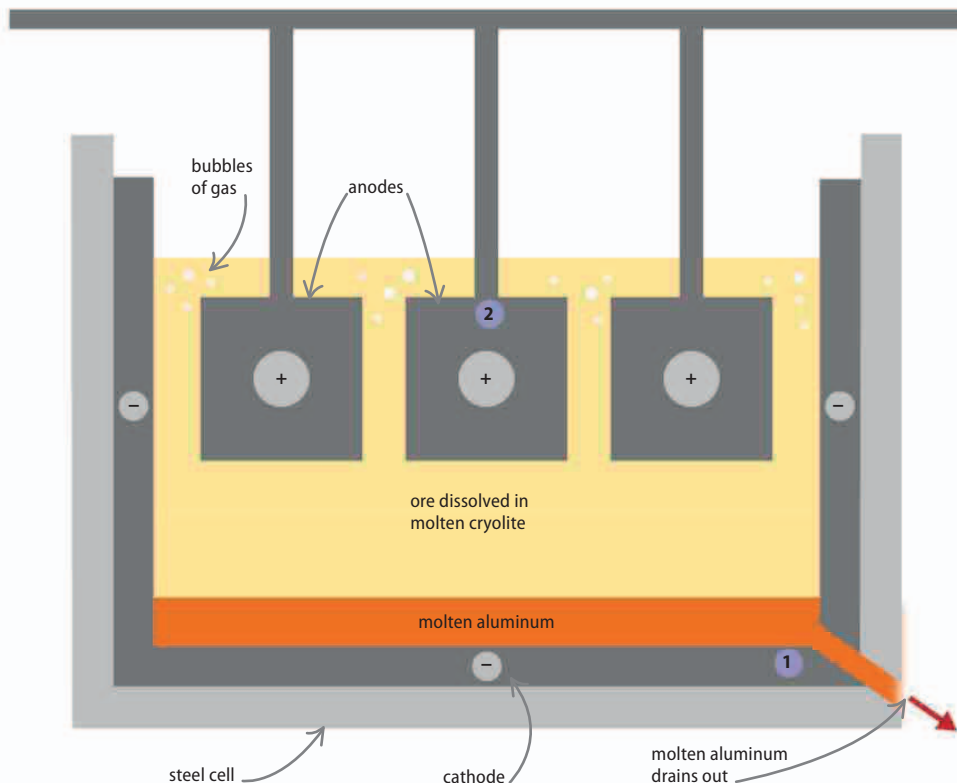
This process is named after Martin Hall and Paul Héroult, who invented it independently of each other in the late 1880s. When an electric current is passed through the electrolyte, it is broken down into positively charged aluminum ions and negatively charged oxygen ions, which are free to move.

1. $\text{Al}^{3+}(\text{l}) + 3\text{e}^- \rightarrow \text{Al}(\text{l})$

Positive aluminum cations are attracted to the negative cathode. Here, each ion (Al^{3+}) receives three electrons from the cathode lining, changing it into an atom of aluminum. The liquid aluminum pools on the cathode at the bottom of the cell and is drained off regularly.

2. $2\text{O}^{2-}(\text{l}) + \text{C}(\text{s}) \rightarrow \text{CO}_2(\text{g}) + 4\text{e}^-$

Negative oxygen ions are attracted to the positive anodes, where each ion loses two electrons to form an oxygen atom. The oxygen reacts with the carbon in the anode to produce carbon dioxide gas, which bubbles out of the liquid. As the carbon is used up, the anodes gradually corrode and need to be replaced.



Alloys

Two or more metals—and sometimes other elements—are mixed together to form an alloy. Alloys exhibit some of the properties of all their individual constituents, so they can be adapted to suit many applications. The first metal implements created by humans were made of bronze, a mixture of copper and tin, two metals that were easy to extract from ores.

COMMON ALLOYS

Name	Main metal	Other metal	Properties	Uses
carbon steel	iron	carbon	high strength	construction
stainless steel	iron	chromium	resistant to corrosion	eating utensils
bronze	copper	tin	easily worked	bells
brass	copper	zinc	does not corrode	zippers, keys
solder	tin	lead	low melting point	soldering
invar	iron	nickel	does not expand when hot	precision
amalgam	mercury	silver	starts soft, then hardens	dental fillings